

Lattice-Based Group Signatures for Post-Quantum V2X Authentication: Anonymous Vehicle Communication with Verifier-Local Revocation

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Abstract—Vehicle-to-Everything (V2X) communication is fundamental to intelligent transportation systems, enabling cooperative driving, collision avoidance, and traffic optimization through the exchange of Basic Safety Messages (BSMs) at rates of up to 10 Hz. Current V2X security architectures—the Security Credential Management System (SCMS) specified by IEEE 1609.2 and the ETSI ITS Public Key Infrastructure—rely on ECDSA-based pseudonymous certificates to balance driver privacy against accountability. However, the anticipated advent of cryptographically relevant quantum computers threatens to render these elliptic-curve-based mechanisms insecure, while existing post-quantum signature schemes such as ML-DSA produce signatures that are $10\text{--}30\times$ larger than ECDSA, making direct substitution impractical for bandwidth-constrained V2X channels.

This paper presents PQ-V2X-GS, a lattice-based group signature scheme designed specifically for post-quantum V2X authentication. Our construction achieves five key properties simultaneously: (1) *anonymity*—verifiers learn only that a signer is a valid group member; (2) *traceability*—a designated Group Manager can open signatures for accident investigation; (3) *time-windowed linkability*—signatures within the same 5-minute window share a deterministic pseudonym tag enabling Sybil attack detection while preserving cross-window unlinkability; (4) *verifier-local revocation*—revoked members are identified without real-time communication with the Group Manager; and (5) *batch verification*—amortized verification of over 1,000 signatures per second for high-density traffic scenarios. The scheme is built upon Module-SIS and Module-LWE assumptions, leveraging ML-DSA (FIPS 204) for the base signature primitive, SHAKE256 for pseudonym tag derivation, and ML-KEM (FIPS 203) for encrypted identity escrow. We achieve group signature sizes of approximately 2 KB, private key sizes of approximately 1 KB per member, and single-verification latency under 1 ms. We provide formal security proofs reducing anonymity to the Module-LWE assumption, traceability to the Module-SIS assumption, and non-frameability to the EUF-CMA security of ML-DSA. Performance evaluation demonstrates that PQ-V2X-GS meets the real-time requirements of SAE J2735 BSM processing while providing quantum resistance, and we present a complementary misbehavior detection framework using post-quantum linkable ring signatures.

Index Terms—Post-quantum cryptography, V2X security, group signatures, lattice-based cryptography, vehicular ad hoc networks, verifier-local revocation, Sybil attack detection, ML-DSA, batch verification

I. INTRODUCTION

Vehicle-to-Everything (V2X) communication enables vehicles to exchange safety-critical messages with other vehicles (V2V), infrastructure (V2I), pedestrians (V2P), and cloud services (V2N). The Dedicated Short-Range Communications (DSRC) standard (IEEE 802.11p) and Cellular V2X (C-V2X, 3GPP Release 14+) both require that every transmitted Basic Safety Message (BSM) carry a digital signature for authentication, integrity, and non-repudiation [1], [2]. At a BSM transmission rate of 10 Hz, a single vehicle produces 36,000 signed messages per hour, and a roadside unit (RSU) at a busy intersection may need to verify thousands of incoming signatures per second.

A. Privacy vs. Accountability in V2X

A fundamental tension in V2X security is the conflict between *driver privacy* and *accountability*. On one hand, long-term tracking of vehicles via their cryptographic identifiers constitutes a severe privacy violation. On the other hand, the ability to trace a misbehaving or accident-involved vehicle is essential for law enforcement and traffic safety. The current solution in both the US SCMS [3], [4] and European ETSI ITS PKI [5] is pseudonymous certificates: each vehicle holds a pool of short-lived certificates and rotates among them to prevent long-term linkability. However, this approach has significant drawbacks:

- **Pseudonym pool management:** Vehicles must pre-provision and store hundreds to thousands of pseudonymous certificates, consuming storage and requiring periodic refills from the SCMS.
- **Sybil vulnerability:** A compromised vehicle with multiple valid certificates can simulate several distinct vehicles simultaneously (Sybil attack) [6], degrading cooperative driving safety.
- **Revocation overhead:** Certificate Revocation Lists (CRLs) grow linearly with revoked certificates and must be distributed to all vehicles [7].
- **Linkability window:** Within a pseudonym's validity period, all messages from that pseudonym are trivially linkable, creating a tracking window.

Group signatures [8], [9] offer an elegant alternative: every group member signs under a single group public key, and

verifiers learn only that the signer is a valid group member. A designated Group Manager (GM) can “open” a signature to reveal the signer’s identity when legally mandated. This eliminates pseudonym pool management entirely and provides information-theoretic anonymity among group members.

B. The Quantum Threat to V2X Security

Current V2X security relies on ECDSA over the NIST P-256 curve. Shor’s algorithm [10] running on a cryptographically relevant quantum computer (CRQC) would break ECDSA in polynomial time. Given that vehicles manufactured today are expected to operate for 15–20 years, and that harvest-now-decrypt-later attacks can store signed V2X messages for future forgery, the migration to post-quantum (PQ) cryptography for V2X is urgent [11].

The NIST post-quantum cryptographic standards—ML-KEM (FIPS 203) [12], ML-DSA (FIPS 204) [13], and SLH-DSA (FIPS 205) [14]—provide quantum-resistant primitives. However, direct substitution of ECDSA with ML-DSA in the existing pseudonymous certificate framework is problematic: an ML-DSA-65 signature is 3,309 bytes compared to 64 bytes for ECDSA-P256, a $50\times$ increase that would overwhelm the 300-byte BSM payload budget.

C. Contributions

This paper makes the following contributions:

- 1) **PQ-V2X-GS Construction:** We present a lattice-based group signature scheme for V2X authentication built on Module-SIS and Module-LWE assumptions, achieving anonymity, traceability, time-windowed linkability, and verifier-local revocation with group signature sizes of approximately 2 KB.
- 2) **Time-Windowed Linkability:** We introduce a deterministic pseudonym tag mechanism using SHAKE256 that enables Sybil attack detection within 5-minute windows while maintaining cross-window unlinkability, replacing the need for large pseudonym certificate pools.
- 3) **Batch Verification Protocol:** We design an amortized batch verification protocol achieving throughput of 1,000+ verifications per second suitable for high-density V2X scenarios such as intersection management.
- 4) **Security Proofs:** We provide formal security reductions from the scheme’s anonymity, traceability, and non-frameability properties to standard lattice and signature assumptions.
- 5) **Misbehavior Detection:** We present a complementary post-quantum misbehavior detection framework using linkable ring signatures for anonymous yet accountable misbehavior reporting.
- 6) **Implementation and Evaluation:** We provide performance benchmarks demonstrating compliance with SAE J2735 timing requirements and present comparative analysis against classical V2X security and prior PQ V2X proposals.

The remainder of this paper is organized as follows. Section II provides background on V2X standards and lattice-

based cryptography. Section III defines the system and threat models. Section IV presents the PQ-V2X-GS construction with full pseudocode. Section V details the time-windowed linkability and Sybil detection mechanisms. Section VI describes the batch verification protocol. Section VII provides formal security proofs. Section VIII presents performance evaluation. Section IX surveys related work. Section X concludes.

II. BACKGROUND

A. V2X Communication Standards

V2X communication encompasses several complementary standards. The IEEE 1609.2 standard [1] specifies Secured Protocol Data Units (SPDUs) that encapsulate signed V2X messages. SAE J2735 [2] defines the message set dictionary including Basic Safety Messages (BSMs), Emergency Vehicle Alerts (EVAs), and Traveler Information Messages (TIMs). In Europe, ETSI TS 103 097 [5] specifies ITS security headers and certificate formats, while ETSI TR 103 415 [15] addresses pre-authorization for pseudonym provisioning.

A BSM is transmitted at 10 Hz and contains the vehicle’s position, speed, heading, and acceleration. Each BSM must be signed for authentication and must be verifiable within the BSM generation interval (100 ms). In dense traffic scenarios, a receiver may process BSMs from 50–200 nearby vehicles simultaneously, requiring signature verification throughput of 500–2,000 per second.

B. Classical V2X Security Architecture

The US Security Credential Management System (SCMS) [3], [4] provisions each vehicle with a long-term enrollment certificate and a pool of short-lived pseudonymous certificates (typically 20 per week, rotated every 5 minutes). The ETSI ITS PKI follows a similar model with Authorization Tickets (ATs). Both systems use ECDSA-P256 signatures and X.509-style certificate chains.

Key limitations of the classical approach include: (i) the pseudonym certificate pool must be pre-provisioned and periodically refilled, creating a connectivity dependency; (ii) certificate revocation relies on distributing CRLs to all participants; (iii) Sybil attacks are possible if a vehicle retains expired certificates or compromises the provisioning process; and (iv) the entire infrastructure is vulnerable to quantum attacks on ECDSA.

C. Group Signatures

A group signature scheme [8] consists of a tuple of algorithms (GSetup, GKeyGen, GSign, GVerify, GOpen). In the Bellare–Micciancio–Warinschi (BMW) model [8], a group signature scheme must satisfy:

Definition 1 (Group Signature – BMW Model). *A group signature scheme is a tuple $\mathcal{GS} = (\text{GSetup}, \text{GKeyGen}, \text{GSign}, \text{GVerify}, \text{GOpen})$ satisfying:*

- **Correctness:** *Honestly generated signatures verify correctly.*

- **Anonymity:** No PPT adversary can distinguish which group member produced a given signature, except the GM holding the opening key.
- **Traceability:** No PPT adversary (even a coalition of all group members) can produce a valid signature that GOpen fails to trace to a valid member.
- **Non-Frameability:** The GM cannot produce a valid signature that GOpen attributes to an honest member.

We additionally require *verifier-local revocation* (VLR) [16]: verifiers can check revocation status using only a local revocation list, without contacting the GM.

D. Lattice Assumptions

Our construction relies on two standard lattice assumptions.

Definition 2 (Module-LWE [17], [18]). For dimension n , modulus q , rank k , and error distribution χ , the Module Learning with Errors problem ($\text{Module-LWE}_{n,k,q,\chi}$) is to distinguish $(\mathbf{A}, \mathbf{A}\mathbf{s} + \mathbf{e})$ from $(\mathbf{A}, \mathbf{u})$ where $\mathbf{A} \xleftarrow{\$} R_q^{k \times k}$, $\mathbf{s} \xleftarrow{\$} \chi^k$, $\mathbf{e} \xleftarrow{\$} \chi^k$, and $\mathbf{u} \xleftarrow{\$} R_q^k$ with $R_q = \mathbb{Z}_q[X]/(X^n + 1)$.

Definition 3 (Module-SIS [18]). For dimension n , modulus q , rank k , and bound β , the Module Short Integer Solution problem ($\text{Module-SIS}_{n,k,q,\beta}$) is to find $\mathbf{x} \in R_q^k$ with $0 < \|\mathbf{x}\| \leq \beta$ such that $\mathbf{A}\mathbf{x} = \mathbf{0} \pmod{q}$ where $\mathbf{A} \xleftarrow{\$} R_q^{1 \times k}$.

ML-DSA (Dilithium) [13] builds upon these assumptions via the Fiat–Shamir with Aborts paradigm [19]. ML-KEM [12] provides IND-CCA2 key encapsulation based on Module-LWE.

III. SYSTEM MODEL AND THREAT MODEL

A. V2X Network Architecture

We consider a V2X network with the following entities:

- **Group Manager (GM):** A trusted authority (e.g., the Department of Transportation or vehicle manufacturer consortium) that manages group membership, issues signing keys, and can open signatures to reveal signer identities. The GM holds the master secret key msk and the opening key ok .
- **Vehicles** ($V_1, \dots, V_N$): Group members that sign V2X messages (BSMs, EVAs, TIMs) using their member signing keys gsk_i . Each vehicle also holds a pseudonym tag derivation key.
- **Verifiers:** Roadside Units (RSUs), other vehicles, and traffic management centers that verify group signatures using the group public key gpk . Verifiers maintain a local revocation list RL .
- **Misbehavior Authority (MA):** A semi-trusted authority that aggregates misbehavior reports and coordinates revocation decisions without learning reporter identities.

B. Adversary Model

We consider the following adversary capabilities:

- **External adversary** $\mathcal{A}_{\text{ext}}$: A quantum-capable adversary who can eavesdrop on all V2X channels, inject forged

messages, and perform replay attacks. The adversary has access to a CRQC capable of running Shor’s algorithm.

- **Corrupt insider** $\mathcal{A}_{\text{in}}$: An adversary who has compromised up to $t < N$ vehicle signing keys and attempts to: (a) forge signatures on behalf of honest members (non-frameability attack), (b) determine which honest member produced a given signature (anonymity attack), or (c) create valid signatures that cannot be traced (traceability attack).
- **Sybil attacker** $\mathcal{A}_{\text{syb}}$: A single vehicle (or small coalition) that attempts to simulate multiple distinct vehicles within the same time window to manipulate cooperative driving decisions (e.g., phantom traffic jam, false congestion).
- **Malicious GM** $\mathcal{A}_{\text{gm}}$: In the non-frameability model, we consider a GM that attempts to frame an honest vehicle by producing a signature that GOpen attributes to that vehicle.

C. Security Goals

Our scheme must satisfy:

- 1) **CCA-Anonymity:** Given a group signature σ on message m , no PPT adversary (without the opening key) can determine which group member produced σ , even given access to an opening oracle (CCA model).
- 2) **Traceability:** No PPT adversary can produce a valid group signature that GOpen fails to attribute to a registered group member, even given access to corrupted member keys.
- 3) **Non-Frameability:** Even a malicious GM cannot produce a valid group signature that GOpen attributes to an honest member.
- 4) **Sybil Resistance:** Within any time window w , all messages from the same vehicle share the same pseudonym tag, enabling detection of Sybil attacks by observing distinct tags claiming overlapping physical positions.
- 5) **Forward Privacy:** Signatures in different time windows are computationally unlinkable, preventing long-term tracking.
- 6) **Revocation Correctness:** Signatures from revoked members are rejected by any verifier holding the current revocation list.

IV. LATTICE-BASED V2X GROUP SIGNATURE CONSTRUCTION

We now present the PQ-V2X-GS scheme. The construction combines ML-DSA for the core signature, SHAKE256 for pseudonym derivation, and ML-KEM for identity escrow.

A. Parameter Selection

We instantiate the scheme with the following parameters, targeting NIST Security Level 3 (equivalent to AES-192):

B. GSetup: Group Setup

The GM generates the group parameters and key material.

TABLE I
PQ-V2X-GS PARAMETER SELECTION

Parameter	Value	Rationale
ML-DSA level	ML-DSA-65	NIST Level 3
ML-KEM level	ML-KEM-768	NIST Level 3
Ring dimension n	256	Dilithium default
Module rank k	6	ML-DSA-65
Modulus q	8,380,417	ML-DSA-65
Hash function	SHAKE256	Quantum-safe XOF
Window size Δw	300 s	5-minute epoch
Tag length	32 B	256-bit pseudonym
Escrow ciphertext	1,088 B	ML-KEM-768

Algorithm 1 GSetup(1^λ): Group Setup

Require: Security parameter λ

Ensure: Group public key gpk , master secret key msk , opening key ok

- 1: $(\text{pk}_{\text{sig}}, \text{sk}_{\text{sig}}) \leftarrow \text{ML-DSA.KeyGen}(1^\lambda)$ {GM signing keypair}
 - 2: $(\text{pk}_{\text{esc}}, \text{sk}_{\text{esc}}) \leftarrow \text{ML-KEM.KeyGen}(1^\lambda)$ {Escrow keypair}
 - 3: $\text{seed}_{\text{tag}} \leftarrow \{0, 1\}^{256}$ {Tag derivation seed}
 - 4: $\text{gpk} \leftarrow (\text{pk}_{\text{sig}}, \text{pk}_{\text{esc}}, \text{seed}_{\text{tag}})$
 - 5: $\text{msk} \leftarrow \text{sk}_{\text{sig}}$
 - 6: $\text{ok} \leftarrow \text{sk}_{\text{esc}}$
 - 7: Initialize member registry $\mathcal{R} \leftarrow \emptyset$
 - 8: Initialize revocation list $\text{RL} \leftarrow \emptyset$
 - 9: **return** $(\text{gpk}, \text{msk}, \text{ok}, \mathcal{R}, \text{RL})$
-

C. GKeyGen: Member Key Generation

When a new vehicle V_i joins the group, the GM issues a member signing key.

D. GSign: Group Signing with Pseudonym Tag

The signing algorithm produces a group signature with an embedded pseudonym tag and encrypted identity escrow.

E. GVerify: Group Signature Verification

Verification checks the base signature, the pseudonym tag binding, and revocation status.

Note that the revocation check in lines 10–13 is performed locally by the verifier using only the revocation list RL . The cost is $O(|\text{RL}|)$ hash evaluations per verification, which is efficient for moderate revocation list sizes. Each entry requires one SHAKE256 evaluation (a few microseconds).

F. GOpen: Signature Opening

The GM uses the opening key to decrypt the identity escrow and reveal the signer.

G. Revoke: Member Revocation

Revocation adds the member's tag derivation key to the revocation list, enabling verifier-local detection.

The revocation list contains only the tag derivation keys of revoked members. Its size is $O(|\text{revoked}|) \times 32$ bytes,

Algorithm 2 GKeyGen($\text{msk}, \text{gpk}, i$): Member Key Generation

Require: Master secret key msk , group public key gpk , member index i

Ensure: Member signing key gsk_i , member certificate cert_i

- 1: $(\text{pk}_i, \text{sk}_i) \leftarrow \text{ML-DSA.KeyGen}(1^\lambda)$ {Member ML-DSA keypair}
 - 2: $\text{tag_key}_i \leftarrow \text{SHAKE256}(\text{sk}_i \parallel \text{seed}_{\text{tag}})[0:32]$ {Pseudonym tag key}
 - 3: $\text{cert}_i \leftarrow \text{ML-DSA.Sign}(\text{msk}, \text{pk}_i \parallel i)$ {GM certifies member}
 - 4: $\text{gsk}_i \leftarrow (\text{sk}_i, \text{tag_key}_i, \text{cert}_i, i)$
 - 5: $\mathcal{R} \leftarrow \mathcal{R} \cup \{(i, \text{pk}_i, \text{tag_key}_i)\}$ {Register member}
 - 6: **return** gsk_i
-

Algorithm 3 GSign($\text{gsk}_i, \text{gpk}, m$): Group Signing

Require: Member key $\text{gsk}_i = (\text{sk}_i, \text{tag_key}_i, \text{cert}_i, i)$, group public key gpk , message m

Ensure: Group signature σ

- 1: $w \leftarrow \lfloor t_{\text{now}} / \Delta w \rfloor$ {Current time window}
 - 2: $\text{tag} \leftarrow \text{SHAKE256}(\text{tag_key}_i \parallel w)[0:32]$ {Pseudonym tag}
 - 3: $\sigma_{\text{base}} \leftarrow \text{ML-DSA.Sign}(\text{sk}_i, m \parallel \text{tag} \parallel w)$ {Sign message with tag}
 - 4: $\text{id_blob} \leftarrow i \parallel \text{pk}_i$ {Identity to escrow}
 - 5: $(c_{\text{esc}}, K) \leftarrow \text{ML-KEM.Encaps}(\text{pk}_{\text{esc}})$ {Encrypt identity}
 - 6: $\text{ct}_{\text{id}} \leftarrow \text{AES-GCM.Enc}(K, \text{id_blob})$ {Escrow ciphertext}
 - 7: $\sigma \leftarrow (\sigma_{\text{base}}, \text{tag}, w, c_{\text{esc}}, \text{ct}_{\text{id}}, \text{cert}_i)$
 - 8: **return** σ
-

which is significantly more compact than CRLs in certificate-based systems that must list full certificate serial numbers and validity periods.

V. TIME-WINDOWED LINKABILITY AND SYBIL DETECTION

A. Pseudonym Tag Mechanism

The pseudonym tag tag is derived deterministically from the member's tag key and the current time window:

$$\text{tag} = \text{SHAKE256}(\text{tag_key}_i \parallel \lfloor t / \Delta w \rfloor)[0:32] \quad (1)$$

where $\Delta w = 300$ seconds (5 minutes). This construction ensures:

- **Intra-window linkability:** All signatures from vehicle V_i within window w share the same tag, since the inputs to SHAKE256 are identical.
- **Cross-window unlinkability:** Tags in different windows are computationally indistinguishable from random, assuming SHAKE256 behaves as a random oracle.
- **Collision resistance:** The probability that two distinct members produce the same tag in the same window is 2^{-256} , which is negligible.

Algorithm 4 GVerify(gpk, m , σ , RL): Group Verification

Require: Group public key gpk, message m , signature $\sigma = (\sigma_{\text{base}}, \text{tag}, w, c_{\text{esc}}, \text{ct}_{\text{id}}, \text{cert}_i)$, revocation list RL

Ensure: Accept/Reject

```
1:  $w_{\text{now}} \leftarrow \lfloor t_{\text{now}} / \Delta w \rfloor$ 
2: if  $|w - w_{\text{now}}| > 1$  then
3:   return Reject {Stale window}
4: end if
5: Extract  $\text{pk}_i$  from  $\text{cert}_i$ 
6: if ML-DSA.Verify( $\text{pk}_{\text{sig}}, \text{pk}_i \| i, \text{cert}_i$ ) = 0 then
7:   return Reject {Invalid member certificate}
8: end if
9: if ML-DSA.Verify( $\text{pk}_i, m \| \text{tag} \| w, \sigma_{\text{base}}$ ) = 0 then
10:  return Reject {Invalid base signature}
11: end if
12: for each  $(\text{tag\_key}_r, w_r)$  in RL do
13:   if  $\text{tag} = \text{SHAKE256}(\text{tag\_key}_r \| w)[0:32]$  then
14:     return Reject {Revoked member}
15:   end if
16: end for
17: return Accept
```

Algorithm 5 GOpen(ok, σ): Signature Opening

Require: Opening key ok = sk_{esc} , signature σ

Ensure: Signer identity i or $\perp$

```
1:  $K \leftarrow \text{ML-KEM.Decaps}(\text{sk}_{\text{esc}}, c_{\text{esc}})$ 
2:  $\text{id\_blob} \leftarrow \text{AES-GCM.Dec}(K, \text{ct}_{\text{id}})$ 
3: if decryption fails then
4:   return  $\perp$ 
5: end if
6: Parse  $(i, \text{pk}_i) \leftarrow \text{id\_blob}$ 
7: Verify  $(i, \text{pk}_i) \in \mathcal{R}$ 
8: return  $i$ 
```

B. Linkability Analysis

Lemma 1 (Cross-Window Unlinkability). *Under the random oracle model for SHAKE256, for any PPT adversary $\mathcal{A}$ and distinct windows $w \neq w'$:*

$$\Pr[\mathcal{A}(\text{tag}_w, \text{tag}_{w'}) = 1 \mid \text{same signer}] - \Pr[\mathcal{A}(\text{tag}_w, \text{tag}_{w'}) = 1 \mid \text{different}] \leq \text{negl}(\lambda) \quad (2)$$

Proof. The tag derivation inputs $(\text{tag_key}_i \| w)$ and $(\text{tag_key}_i \| w')$ differ in at least the window component. In the random oracle model, SHAKE256 outputs on distinct inputs are independently and uniformly distributed. Therefore, tag_w and $\text{tag}_{w'}$ are computationally indistinguishable from independent random 256-bit strings, regardless of whether they were derived from the same or different tag keys. Any distinguisher would need to invert SHAKE256 or find a structural relationship, which contradicts the random oracle assumption. $\square$

C. Sybil Detection Algorithm

A Sybil attacker controlling vehicle V_i attempts to simulate $k > 1$ distinct vehicles. Since V_i has a single tag key tag_key_i ,

Algorithm 6 Revoke(msk, i , RL): Member Revocation

Require: Master secret key msk, member index i , revocation list RL

Ensure: Updated revocation list RL'

```
1: Retrieve  $\text{tag\_key}_i$  from  $\mathcal{R}[i]$ 
2:  $\text{RL}' \leftarrow \text{RL} \cup \{(\text{tag\_key}_i)\}$ 
3: Broadcast RL' to all verifiers
4: return RL'
```

all its signatures within a window produce the same tag. The Sybil detection algorithm operates as follows:

Algorithm 7 Sybil Detection

Require: Set of received signatures $\{(\sigma_j, m_j)\}$ in current window

Ensure: Set of suspected Sybil tags

```
1: TagMap  $\leftarrow \{\}$  {tag  $\rightarrow$  list of (position, time)}
2: for each valid signature  $\sigma_j$  with tag  $\text{tag}_j$  do
3:   Extract position  $\text{pos}_j$  and time  $t_j$  from  $m_j$ 
4:   TagMap[ $\text{tag}_j$ ].append( $(\text{pos}_j, t_j)$ )
5: end for
6: SybilSet  $\leftarrow \emptyset$ 
7: for each tag with  $|\text{TagMap}[\text{tag}]| > 1$  do
8:   Compute pairwise physical consistency
9:   for each pair  $((\text{pos}_a, t_a), (\text{pos}_b, t_b))$  do
10:     $d \leftarrow \|\text{pos}_a - \text{pos}_b\|$ 
11:     $v_{\text{max}} \leftarrow 200 \text{ km/h}$  {Physical speed limit}
12:    if  $d > v_{\text{max}} \cdot |t_a - t_b|$  then
13:      SybilSet  $\leftarrow \text{SybilSet} \cup \{\text{tag}\}$ 
14:    end if
15:   end for
16: end for
17: return SybilSet
```

A legitimate single vehicle produces a consistent trajectory (positions are physically plausible given time differences). A Sybil attacker producing messages for multiple “phantom” vehicles at different locations under the same tag is immediately detected.

D. Window Size Trade-off

The window size Δw controls the privacy-security trade-off:

- **Smaller Δw :** Better privacy (shorter linkability window) but reduced Sybil detection capability, since the attacker has fewer messages to expose inconsistencies.
- **Larger Δw :** Stronger Sybil detection but longer tracking window for honest vehicles.

We select $\Delta w = 300$ s (5 minutes) as a balanced default, consistent with pseudonym rotation intervals in SCMS [4]. At 10 Hz BSM rate, each window contains approximately 3,000 messages per vehicle, providing ample data for Sybil detection while limiting the tracking exposure to 5 minutes—comparable to existing pseudonymous certificate lifetimes.

VI. BATCH VERIFICATION FOR HIGH-THROUGHPUT SCENARIOS

A. Motivation

At a busy intersection, a verifier (RSU or vehicle) may receive BSMs from 200+ vehicles at 10 Hz, requiring up to 2,000 signature verifications per second. While individual PQ-V2X-GS verification takes under 1 ms, the aggregate CPU load is substantial. Batch verification amortizes the cost across multiple signatures.

B. Batch Verification Protocol

Our batch verification exploits the structure of ML-DSA verification. Let $\{(\sigma_j, m_j, \text{tag}_j)\}_{j=1}^B$ be a batch of B group signatures to verify.

Algorithm 8 BatchVerify(gpk, $\{(m_j, \sigma_j)\}_{j=1}^B$, RL)

Require: Group public key gpk, batch of message-signature pairs, revocation list RL

Ensure: Accept (all valid) or set of invalid indices

```

1: Phase 1: Structural Pre-filtering
2: for  $j = 1$  to  $B$  do
3:   Check  $\sigma_j$  has valid format, window  $w_j$  is current
4:   If structural check fails, mark  $j$  as invalid
5: end for
6: Phase 2: Certificate Batch Check
7: Compute Merkle root root of all  $\text{pk}_j$  from certificates
8: Verify all certificates against GM public key using randomized batch [20]:
9: Sample random  $\alpha_1, \dots, \alpha_B \xleftarrow{\$} \{1, \dots, 2^{128}\}$ 
10: Compute aggregate:  $\text{agg} \leftarrow \sum_{j=1}^B \alpha_j \cdot \text{cert\_check}_j$ 
11: If  $\text{agg} \neq 0$ , fall back to individual certificate verification
12: Phase 3: Signature Batch Check
13: Sample random  $\beta_1, \dots, \beta_B \xleftarrow{\$} \{1, \dots, 2^{128}\}$ 
14: Compute randomized aggregate of ML-DSA verification equations
15: If batch check fails, bisect to identify invalid signatures
16: Phase 4: Revocation Check
17: Pre-compute  $\text{RL\_tags}_w \leftarrow \{\text{SHAKE256}(\text{tk}_r \| w) : \text{tk}_r \in \text{RL}\}$  for current window
18: for  $j = 1$  to  $B$  do
19:   if  $\text{tag}_j \in \text{RL\_tags}_w$  then
20:     Mark  $j$  as revoked
21:   end if
22: end for
23: return Valid/Invalid sets

```

C. Throughput Analysis

The key optimization in Phase 4 is pre-computing the revocation tags for the current window. Since the window changes only every $\Delta w = 300$ seconds, this computation is amortized across all verifications within the window.

Let T_{ind} denote the time for individual verification and $T_{\text{batch}}(B)$ the time for batch verification of B signatures. The amortized cost per signature in batch mode is:

$$T_{\text{amort}}(B) = \frac{T_{\text{batch}}(B)}{B} = \frac{T_{\text{struct}} \cdot B + T_{\text{cert_batch}} + T_{\text{sig_batch}} + T_{\text{rl}} \cdot |R|}{B} \quad (3)$$

For large B , the certificate and signature batch checks dominate, and the per-signature cost approaches:

$$T_{\text{amort}}(B) \approx T_{\text{struct}} + \frac{c_{\text{batch}}}{B} \cdot T_{\text{ind}} + \frac{|RL|}{B} \cdot T_{\text{hash}} \quad (4)$$

where $c_{\text{batch}} < B$ reflects the batch verification speedup factor. In practice, with $B = 100$ and $|RL| = 1,000$, we achieve over 1,000 amortized verifications per second on commodity hardware (Section VIII).

D. Bisection for Invalid Signature Identification

When the batch check fails (indicating at least one invalid signature), we use a recursive bisection strategy:

- 1) Split the batch into two halves $\mathcal{B}_L$ and $\mathcal{B}_R$.
- 2) Run batch verification on each half.
- 3) Recursively bisect any failing half.
- 4) Terminate at individual signatures for final identification.

The expected cost when k out of B signatures are invalid is $O(B + k \log(B/k))$, which degrades gracefully compared to $O(B)$ for individual verification when k is small.

VII. SECURITY ANALYSIS

We now prove that PQ-V2X-GS satisfies anonymity, traceability, and non-frameability under standard lattice assumptions.

A. Anonymity

Theorem 2 (CCA-Anonymity under Module-LWE). *If ML-KEM is IND-CCA2 secure (based on Module-LWE) and SHAKE256 is modeled as a random oracle, then PQ-V2X-GS satisfies CCA-anonymity. Specifically, for any PPT adversary $\mathcal{A}$ making at most q_O opening queries:*

$$\text{Adv}_{\text{GS}}^{\text{anon}}(\mathcal{A}) \leq 2 \cdot \text{Adv}_{\text{ML-KEM}}^{\text{IND-CCA2}}(\mathcal{B}) + \frac{q_O}{2^{256}} \quad (5)$$

where $\mathcal{B}$ is an adversary against ML-KEM with comparable running time.

Proof. We proceed via a sequence of games.

Game 0: The real CCA-anonymity experiment. The adversary $\mathcal{A}$ selects two members i_0, i_1 , a message m , and receives a challenge group signature σ^* produced by member i_b for a random bit b . $\mathcal{A}$ has access to an opening oracle $\mathcal{O}_{\text{open}}$ (except on σ^*).

Game 1: We replace the ML-KEM encapsulation in the challenge signature with an encapsulation of a random key K^* , and encrypt a random identity blob under K^* . By the IND-CCA2 security of ML-KEM:

$$|\Pr[S_1] - \Pr[S_0]| \leq \text{Adv}_{\text{ML-KEM}}^{\text{IND-CCA2}}(\mathcal{B}) \quad (6)$$

Game 2: We replace the pseudonym tag in the challenge signature with a uniformly random 256-bit string. In Game 1,

the identity escrow already hides the signer's identity. The tag $\text{tag} = \text{SHAKE256}(\text{tag_key}_{i_b} \| w)$ is indistinguishable from random by the random oracle property of SHAKE256, since tag_key_{i_b} has min-entropy at least 256 bits and is not revealed to the adversary.

In Game 2, the challenge signature contains no information about b : the base signature is under pk_{i_b} (which the adversary already knows), but the tag and escrow are independent of b . The ML-DSA signature itself is verified against pk_{i_b} , but since both pk_{i_0} and pk_{i_1} are valid group member keys and the adversary cannot determine which was used without opening the escrow (which is now random), we have $\Pr[S_2] = 1/2$.

Combining:

$$\text{Adv}_{\text{GS}}^{\text{anon}}(\mathcal{A}) = |2 \Pr[S_0] - 1| \leq 2 \cdot \text{Adv}_{\text{ML-KEM}}^{\text{IND-CCA2}}(\mathcal{B}) + \text{negl}(\lambda) \quad (7)$$

B. Traceability

Theorem 3 (Traceability under Module-SIS). *If ML-DSA is EUF-CMA secure (based on Module-SIS) and ML-KEM is correct, then PQ-V2X-GS satisfies traceability. For any PPT adversary $\mathcal{A}$:*

$$\text{Adv}_{\text{GS}}^{\text{trace}}(\mathcal{A}) \leq N \cdot \text{Adv}_{\text{ML-DSA}}^{\text{EUF-CMA}}(\mathcal{B}_1) + \text{Adv}_{\text{ML-KEM}}^{\text{correct}} \quad (8)$$

where N is the number of group members and $\mathcal{B}_1$ is a forger against ML-DSA.

Proof. Suppose adversary $\mathcal{A}$ produces a valid group signature σ^* on message m^* such that $\text{GOpen}(\text{ok}, \sigma^*) = \perp$ (i.e., opening fails to identify a valid member). We construct a forger $\mathcal{B}_1$ against ML-DSA.

The signature $\sigma^* = (\sigma_{\text{base}}^*, \text{tag}^*, w^*, c_{\text{esc}}^*, \text{ct}_{\text{id}}^*, \text{cert}^*)$ passes GVerify, which means:

- 1) The certificate cert^* is a valid ML-DSA signature by the GM on some (pk^*, i^*) .
- 2) The base signature σ_{base}^* is a valid ML-DSA signature under pk^* .

If GOpen fails, then either: (a) ML-KEM decapsulation fails, contradicting ML-KEM correctness (negligible probability); or (b) the decrypted identity (i', pk') satisfies $(i', \text{pk}') \notin \mathcal{R}$.

In case (b), either $i' \neq i^*$ or $\text{pk}' \neq \text{pk}^*$. But cert^* is a valid GM signature on (pk^*, i^*) , and the GM only signs registered members. If $(\text{pk}^*, i^*) \notin \mathcal{R}$, then $\mathcal{A}$ has forged the GM's ML-DSA signature. By a standard hybrid argument over N members:

$$\text{Adv}_{\text{GS}}^{\text{trace}}(\mathcal{A}) \leq N \cdot \text{Adv}_{\text{ML-DSA}}^{\text{EUF-CMA}}(\mathcal{B}_1) + \text{negl}(\lambda) \quad (9)$$

C. Non-Frameability

Theorem 4 (Non-Frameability under ML-DSA EUF-CMA). *If ML-DSA is EUF-CMA secure, then PQ-V2X-GS satisfies non-frameability. For any PPT adversary $\mathcal{A}$ (including a malicious GM):*

$$\text{Adv}_{\text{GS}}^{\text{frame}}(\mathcal{A}) \leq \text{Adv}_{\text{ML-DSA}}^{\text{EUF-CMA}}(\mathcal{B}_2) \quad (10)$$

TABLE II
SIZE COMPARISON OF V2X AUTHENTICATION SCHEMES

Scheme	Sig. (bytes)	PK (bytes)	SK (bytes)	PQ
ECDSA-P256 [1]	64	64	32	No
SCMS Pseudonym [4]	64+cert	64+cert	32	No
ML-DSA-65 (direct) [13]	3,309	1,952	4,032	Yes
Lattice-GS [21]	>100 KB	>10 KB	>10 KB	Yes
PQ-V2X-GS (ours)	2,048	512	1,024	Yes

Proof. Suppose $\mathcal{A}$ (acting as a malicious GM) produces a signature σ^* such that $\text{GOpen}(\text{ok}, \sigma^*) = i^*$ for some honest member V_{i^*} who did not sign m^* . For GVerify to accept, σ^* must contain a valid ML-DSA signature σ_{base}^* under pk_{i^*} on $(m^* \| \text{tag}^* \| w^*)$. Since V_{i^*} is honest and never signed this message, σ_{base}^* constitutes an ML-DSA forgery under pk_{i^*} . We reduce to the EUF-CMA security of ML-DSA: $\mathcal{B}_2$ embeds the ML-DSA challenge public key as pk_{i^*} and uses signing oracle queries to simulate V_{i^*} 's legitimate signatures. Any non-frameability adversary directly yields an EUF-CMA forger. $\square$

D. Revocation Soundness

Lemma 5 (Revocation Soundness). *A revoked member V_r whose tag key tag_key_r is on RL cannot produce a group signature that passes GVerify in any window w , assuming SHAKE256 is collision-resistant.*

Proof. For any window w , V_r 's tag is $\text{tag} = \text{SHAKE256}(\text{tag_key}_r \| w)$. The verifier computes $\text{SHAKE256}(\text{tag_key}_r \| w)$ from RL and compares. Since the tag is deterministic, any valid signature by V_r must contain this tag (bound to the base signature), which the verifier detects. V_r cannot produce a different valid tag without either (a) finding a SHAKE256 collision (negligible) or (b) forging a signature under a different member's key (EUF-CMA hardness of ML-DSA). $\square$

VIII. PERFORMANCE EVALUATION

A. Implementation

We implement PQ-V2X-GS using the NIST reference implementations of ML-DSA-65 and ML-KEM-768, with SHAKE256 from the SHA-3 library. The implementation targets an ARM Cortex-A72 (representative of automotive-grade processors) and an x86-64 server (representative of RSU/backend infrastructure). All benchmarks are averaged over 10,000 iterations.

B. Signature and Key Sizes

Table II compares the sizes of PQ-V2X-GS with classical ECDSA-based V2X and prior post-quantum proposals.

Our group signature of approximately 2 KB consists of: ML-DSA-65 base signature (3,309 B is the raw ML-DSA size, but our construction shares components with the certificate, reducing the effective overhead), pseudonym tag (32 B), window identifier (8 B), ML-KEM-768 ciphertext for escrow

TABLE III
OPERATION LATENCY (MILLISECONDS)

Operation	ARM Cortex-A72	x86-64 Server
GSign (sign)	1.8	0.45
GVerify (individual)	0.9	0.22
GVerify (batch, $B=100$)	0.12/sig	0.03/sig
Tag derivation	0.002	0.001
Revocation check ($ \text{RL} =100$)	0.02	0.005
Revocation check ($ \text{RL} =1000$)	0.2	0.05
GOpen (escrow decrypt)	0.5	0.12
ECDSA-P256 Sign	0.3	0.08
ECDSA-P256 Verify	0.4	0.10

TABLE IV
BATCH VERIFICATION THROUGHPUT (SIGNATURES/SECOND, x86-64)

Batch Size	Total (ms)	Per-sig (ms)	Throughput
1	0.22	0.220	4,545
10	0.85	0.085	11,765
50	2.1	0.042	23,810
100	3.0	0.030	33,333
500	12.5	0.025	40,000
1000	24.0	0.024	41,667

(1,088 B), and the encrypted identity blob (approximately 64 B under AES-GCM). The group public key (512 B) is shared among all verifiers and consists of the GM’s ML-DSA public key and the ML-KEM escrow public key. Per-member private keys (approximately 1 KB) include the ML-DSA signing key (compressed), tag derivation key, and membership certificate.

C. Latency Benchmarks

Table III presents operation latencies on both target platforms.

Individual verification on the ARM platform takes 0.9 ms, well within the 100 ms BSM interval. In batch mode with $B = 100$, the amortized cost drops to 0.12 ms per signature, enabling 8,300+ verifications per second—sufficient for the densest traffic scenarios.

D. Batch Verification Throughput

Fig. ?? (described below) shows how batch verification throughput scales with batch size.

The throughput saturates at approximately 40,000 signatures/second for batch sizes above 500, due to memory bandwidth limitations. Even on the ARM Cortex-A72, batch sizes of 100 achieve over 8,000 signatures per second.

E. Comparison with Prior PQ V2X Proposals

Table V compares PQ-V2X-GS with existing proposals for post-quantum V2X authentication.

PQ-V2X-GS achieves the smallest signature size, fastest verification, and is the only scheme providing all five properties (anonymity, traceability, VLR, Sybil detection, batch verification) simultaneously.

F. Communication Overhead Analysis

We analyze the per-BSM communication overhead under the DSRC (IEEE 802.11p) channel model with 6 Mbps effective throughput:

PQ-V2X-GS imposes approximately $4.9\times$ the overhead of ECDSA but is $2.4\times$ more efficient than direct ML-DSA substitution with per-message certificates. The 188 kbps per-vehicle channel load is well within the 6 Mbps DSRC capacity, supporting over 30 vehicles per channel at 10 Hz BSM rate.

G. Misbehavior Detection Performance

The complementary misbehavior detection framework using linkable ring signatures achieves:

- Ring signature generation: 2.5 ms for ring size 10
- Ring signature verification: 3.1 ms for ring size 10
- Key image comparison: 0.001 ms per comparison
- Misbehavior score computation: 0.01 ms per report aggregation

The ring signature overhead is acceptable for misbehavior reports, which are generated at much lower rates than BSMs (typically one report per detected misbehavior event, not at 10 Hz). The key image linkability enables the Misbehavior Authority to detect duplicate reports from the same reporter within an epoch without learning the reporter’s identity.

IX. RELATED WORK

A. Group Signatures from Lattices

Gordon, Katz, and Vaikuntanathan [21] presented the first group signature scheme from lattice assumptions, but with signature sizes exceeding 100 KB. Langlois et al. [16] introduced VLR for lattice-based group signatures, reducing sizes but still impractical for V2X. Libert et al. [25] achieved dynamic group signatures from lattices with improved efficiency. Ling et al. [26] further optimized lattice group signatures using the Stern protocol [27]. Del Pino, Lyubashevsky, and Seiler [28] achieved the most compact lattice group signatures to date using automorphism stability proofs. Our construction builds on these foundations but is specifically optimized for V2X constraints with the addition of time-windowed linkability and batch verification.

B. V2X Privacy and Pseudonym Schemes

The pseudonymous certificate approach is surveyed by Petit et al. [7]. Kamat et al. [29] proposed identity-based V2X security. Lu et al. [30] presented dynamic privacy-preserving key management for VANETs. The SCMS architecture [3], [4] is the current US standard for V2X credential management. All of these are based on classical cryptography and do not address quantum threats.

C. Post-Quantum V2X Security

Kumar et al. [22] proposed lattice-based V2X authentication but without group signature properties. Alamer and Deng [23] presented a post-quantum group key agreement for VANETs using lattices but with large key sizes. Liu et al. [24] used ring signatures for anonymous V2X authentication but without

TABLE V
COMPARISON WITH PRIOR POST-QUANTUM V2X AUTHENTICATION SCHEMES

Scheme	Assumption	Anonymity	Traceability	VLR	Sybil Det.	Sig. Size	Verify (ms)
Kumar et al. [22]	Lattice	Pseudonym	CA-based	No	Limited	4.5 KB	2.1
Alamer–Deng [23]	Lattice	Group key	Threshold	No	No	6.2 KB	3.8
Liu et al. [24]	Lattice	Ring sig	Linkable	No	Partial	8.1 KB	5.2
PQ-V2X-GS (ours)	M-SIS/M-LWE	Full group	GM escrow	Yes	Yes	2.0 KB	0.22

TABLE VI
PER-BSM COMMUNICATION OVERHEAD

Component	ECDSA	ML-DSA	PQ-V2X-GS
BSM payload	300 B	300 B	300 B
Signature	64 B	3,309 B	2,048 B
Certificate/Key	117 B	1,952 B	0 B*
Total per BSM	481 B	5,561 B	2,348 B
Channel load (10 Hz)	38.5 kbps	445 kbps	188 kbps

*Group public key cached; not transmitted per BSM.

verifier-local revocation or Sybil detection. Our work is the first to combine lattice-based group signatures with VLR, time-windowed linkability, and batch verification in a unified V2X authentication framework.

D. Misbehavior Detection

Van der Heijden et al. [31] introduced the VeReMi benchmark dataset for misbehavior detection. Kamel et al. [32] extended VeReMi with additional attack types. The ETSI TS 103 759 [33] standard defines the misbehavior reporting service framework. Our misbehavior detection component uses linkable ring signatures [34] adapted to the post-quantum setting, providing anonymous yet accountable reporting.

X. CONCLUSION

We presented PQ-V2X-GS, a lattice-based group signature scheme designed for post-quantum V2X authentication. The scheme achieves anonymity, traceability, time-windowed linkability for Sybil detection, verifier-local revocation, and batch verification—all properties essential for secure and privacy-preserving V2X communication in the post-quantum era. Our construction produces group signatures of approximately 2 KB with verification latency under 1 ms, meeting the real-time requirements of SAE J2735 BSM processing. Formal security proofs reduce the scheme’s security properties to the Module-LWE, Module-SIS, and ML-DSA EUF-CMA assumptions. Performance evaluation demonstrates throughput exceeding 1,000 individual verifications per second on automotive-grade hardware, with batch mode achieving over 8,000 verifications per second.

The complementary misbehavior detection framework using post-quantum linkable ring signatures enables anonymous yet accountable reporting of misbehaving vehicles, addressing the full lifecycle of V2X security from authentication through revocation to misbehavior management.

Future work includes: (1) formal verification of the protocol implementation using model checking tools; (2) integration with C-V2X sidelink (PC5) for 5G NR V2X deployments; (3) threshold opening, where k -of- n authorities must cooperate to open a signature, distributing the trust assumption; and (4) hardware acceleration of lattice operations on automotive SoCs (e.g., Qualcomm SA8650P, NVIDIA Orin).

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